SafeBite Mobile Application

UI/UX Screen Design and Implementation Reference

Prepared from SafeBite – FYP SRS Document V4

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1 Purpose and design baseline

This guide translates SafeBite SRS V4 into a practical screen plan for Figma or Google Stitch. It is a design reference, not a replacement for the SRS. SafeBite is a food-safety decision-support application for Pakistani packaged foods.

1.1 Non-negotiable rules

- Missing values are never zero and never evidence of safety.
- Research-derived values show evidence level: product-specific, brand-level, or category-level.
- CLR, CCI, and Decision-Support Grade are concentration-comparison outputs, not personal exposure estimates, diagnoses, risk assessments, compliance certifications, or safety guarantees.
- Use exact status text and reason codes from the SRS status catalogue.
- Only eligible product-specific evidence may support CCI, grades, or automatic alternative recommendations as defined by the SRS.
- Health-context tags are optional, require explicit consent, and may only show cited informational context. Never label a product safe/unsafe for a person.
- Exposure-reduction guidance must come from curated, sourced, reviewed records. Do not claim detox, exercise, sweating, supplements, or diets remove metals.
- Cached results show cache status and retrieval time. TTL is seven days; expired results must not appear current.

2 Design system

Token	Starting specification
Target frame	Android portrait 393 × 852 px; verify other sizes
Primary green	`#176B4D`
Background	`#F7F8F5`
Main text	`#202923`
Secondary text	`#66716A`
Card surface	`#FFFFFF`
Border	`#E1E7E1`
Spacing	8 px base scale
Card radius	12–20 px
Button height	48–54 px minimum target
Typography	Inter or similar readable sans-serif

These are recommendations, not SRS-mandated values. Use text and icons as well as colour to communicate status.

2.1 Primary navigation

Use five signed-in destinations: Home, Scan, Ask AI, Saved, Profile. Saved contains Favorites and Watchlist. History should be reachable within two selections. Keep the scan/search-to-interpretation path short.

3 Screen-by-screen specifications

Generate one screen at a time, reuse the same design system, and check every label manually. Prompts below can be pasted into Figma AI or Google Stitch.

3.1 1. Splash

Layout: centered leaf/food logo, SafeBite wordmark, tagline “Understand what’s in your food.”, subtle loading indicator. No buttons. Route authenticated users to Home, others to Welcome. **Prompt:** Design a minimalist SafeBite Android splash screen, 393 x 852 px, forest green #176B4D and warm off-white #F7F8F5. Center a leaf-and-checkmark logo, SafeBite wordmark, tagline “Understand what’s in your food.” and subtle loading indicator. No buttons or clutter.

3.2 2. Welcome

Layout: logo, friendly packaged-food/barcode illustration, heading “Make more informed food choices”, brief copy, primary Create account, secondary Sign in, decision-support disclaimer. **Prompt:** Create a polished SafeBite welcome screen for Android 393 x 852 px. Forest-green/off-white theme, packaged groceries and barcode illustration, heading “Make more informed food choices”, concise supporting text, full-width “Create account”, outlined “Sign in”, and readable disclaimer that SafeBite does not guarantee safety or provide medical diagnosis.

3.3 3. Create account

Layout: back button, title, fields required by the agreed API contract, show/hide password, consent/terms if applicable, Create account button, Sign in link. **States:** invalid input, duplicate account, network failure, loading, success. **Prompt:** Design a SafeBite registration screen with back arrow, “Create your account”, labelled email/password fields and only other fields required by the final API contract, password visibility control, inline validation, primary Create account button, and Sign in link. Do not invent social login.

3.4 4. Sign in

Layout: logo, Welcome back, email/password, visibility control, Sign in, Create account link; Forgot password only if implemented. **Prompt:** Create a clean SafeBite sign-in screen with logo, “Welcome back”, labelled email/password fields, password visibility icon, green Sign in button, Create account link, and space for inline errors. Include password recovery only if supported.

3.5 5. Home dashboard

Layout: greeting/profile icon; search field for product, brand, category, barcode; dominant Scan a product card; Search products shortcut; recent items; Ask SafeBite AI banner; optional health-context setup; bottom navigation. **Prompt:** Create a SafeBite Home dashboard, 393 x 852 px, with greeting, profile icon, rounded product search field, prominent Scan a product card, secondary search action, recent product cards, Ask SafeBite AI banner, optional health-context setup card, and bottom navigation Home/Scan/Ask AI/Saved/Profile. Scanning must be the clearest action.

3.6 6. Barcode scanner

Layout: camera preview, close/back, title, scan frame, alignment instruction, flashlight, manual barcode entry, permission-denied guidance. **States:** permission request/denied, scanning, detected, not found, camera unavailable, network failure. **Prompt:** Design a SafeBite scanner screen with dark camera preview, close icon, “Scan product barcode”, centered scan frame with green corner guides, “Align the barcode inside the frame”, flashlight toggle, and “Enter barcode manually”. Include a permission-denied state. Do not show fake results.

3.7 7. Product search

Layout: search and clear controls, recent queries, supported filters, results count, product cards with image/name/brand/category/barcode when available. **States:** loading, no results, failure, populated results. **Prompt:** Create a SafeBite product-search screen with back arrow, search input, recent searches, supported filter chips, product cards with image/name/brand/category, and "No matching products found" empty state with retry. Use consistent green/off-white styling.

3.8 8. Product details

Layout: image, name, brand, category, barcode; favorite/watchlist; analysis summary; evidence-level label; contaminant summary; View full analysis, Compare, Ask AI, View sources. **Rule:** show exact SRS status/reason for missing data or incomparable results. Never replace missing evidence with a reassuring grade. **Prompt:** Design SafeBite product details with product identity, favorite/watchlist icons, analysis summary with exact status, reason and evidence level, available contaminant rows, and actions View full analysis, Compare product, Ask SafeBite AI, View research sources. Include an insufficient-data state; missing evidence must not imply safety.

3.9 9. Full contaminant analysis

Layout: product name and timestamp; grade only when eligible; CLR, normalized margin, CCI cards with explanations; contaminant rows with measured value/unit, reference limit/unit, exact status, reason code, evidence level and source; expandable interpretation, exclusions and rule version. **Prompt:** Create an approachable scientific analysis screen with product header and timestamp, eligible grade or neutral unavailable state, CLR/normalized-margin/CCI cards, contaminant rows showing value/unit, limit/unit, exact status, reason, evidence level and source, plus expandable interpretation/calculation details. Do not invent thresholds or imply diagnosis.

3.10 10. Heavy-metal interpretation

Layout: exact comparison status and reason; contaminant/value/unit; applicable limit/unit; why the reference applies; what evidence establishes and does not establish; source link. **Prompt:** Design a plain-language SafeBite interpretation screen with contaminant, measured value/unit, applicable limit/unit, exact status/reason, evidence level, selected source and concise explanation. State that concentration comparison is not personal exposure or diagnosis.

3.11 11. Research evidence and sources

Layout: evidence level; source cards with title, authors, year, journal, location, sample size, method, URL/DOI when available; product relationship and limitations; open-source action. **Prompt:** Create a SafeBite Research and Evidence screen with evidence level and source cards showing title, authors, year, journal, location/sample size/method when available, product relationship (product-specific, brand-level, category-level), limitations, and Open source. Include a no-evidence state; do not imply category research tested the exact product.

3.12 12. Alternatives

Layout: eligibility explanation; alternatives with name/brand, comparable evidence-backed results, source/evidence labels, View product action. **Rule:** automatic recommendations require eligible product-specific evidence. If none qualify, say so rather than ranking unsupported products. **Prompt:** Design an evidence-based alternatives screen with explanation of eligibility, product cards containing only comparable evidence-backed results and source labels, and View product actions. Include "No eligible alternatives found". Do not invent winners.

3.13 13. Product comparison

Layout: product cards, add/remove, aligned rows for contaminant/value/unit/reference limit/status/evidence level, incompatibility warnings. **Prompt:** Create a comparison screen with two product cards, Add product, and aligned comparison rows for contaminant, value/unit, reference limit, status and evidence level. Mark missing/incomparable results and explain why. Never fill missing values with zero or declare a winner without comparable evidence.

3.14 14. AI Food Safety Assistant

Layout: header and optional product context; suggested questions; chat bubbles; citation cards; composer/send; loading state. **States:** insufficient retrieval, network/provider failure, retry. Use the SRS label "No Supporting Evidence Found" where applicable. **Prompt:** Design a citation-grounded SafeBite AI chat with product context, suggested questions, chat bubbles, source citations with title/year and open-source action, composer, loading state and "No Supporting Evidence Found" state. No fabricated citations or individualized medical advice.

3.15 15. Search and scan history

Layout: Scans/Searches tabs; chronological product/action/date rows; reopen product; clear-history confirmation; empty state. **Prompt:** Create SafeBite history with Scans and Searches tabs, chronological product rows with thumbnail/name/action/date, clear-history confirmation and friendly empty state. Keep reachable within two selections.

3.16 16. Favorites and Watchlist

Layout: tabs; product cards; remove action; updated-evidence indicator only when verified; empty states. **Prompt:** Design Saved Products with Favorites and Watchlist tabs, product cards, remove actions, verified updated-evidence indicator, and an empty state with Scan a product action.

3.17 17. Profile and settings

Layout: account name/email; edit profile; health-context preferences; history; privacy/data; supported settings; about; sign out and confirmation. **Prompt:** Create SafeBite Profile Settings with avatar/name/email, edit profile, health-context preferences, history, favorites/watchlist, privacy/data, about and sign out. Use grouped rows and a sign-out confirmation. Do not expose health tags on Home.

3.18 18. Health-context consent and tag management

Layout: explain purpose and limitations; explicit unchecked consent; approved optional tags; Save, Skip, Edit, Delete; privacy note and delete confirmation. **Prompt:** Design a privacy-first health-context consent screen. Explain that tags are optional and used for curated cited research context, not exposure estimation or personal safety verdicts. Include unchecked explicit consent, approved tag options, Save, Skip, Edit, Delete and deletion confirmation. Never pre-check consent.

3.19 19. Health-context interpretation

Layout: for each measured contaminant and selected tag, show deterministically matched reviewed evidence: summary, evidence strength, citation, reviewer and review date. State that CLR is not the user's exposure. **States:** exact "No Health-Context Evidence Found" when no match; doctor/dietitian referral, stronger for pregnancy, young children or "Above Limit". **Prompt:** Design a health-context interpretation section with contaminant, comparison status, reviewed note matched to a user tag, evidence strength, citation, reviewer/date, statement that CLR is not the user's

exposure, and doctor/dietitian advice. Show “No Health-Context Evidence Found” when no match. Never label a product safe/unsafe for the user.

3.20 20. Exposure-reduction guidance

Layout: show only for eligible “Close to Limit”, “At Limit” or “Above Limit” results; curated general guidance, source, reviewer/date, consider-alternatives action and consultation advice. **Do not show guidance** for “Not Tested”, “No Applicable Limit Found” or “Not Comparable” beyond saying no conclusion is possible. Do not claim detox, sweating, exercise, supplements or diets remove metals. **Prompt:** Design a SafeBite guidance section shown only for eligible Close to Limit/At Limit/Above Limit statuses. Show general reviewed guidance, citation, reviewer/date, alternatives action and professional consultation advice. For unavailable/incomparable results state no conclusion is possible. No detox claims.

3.21 21. Offline and cache states

Layout: exact cache status, retrieval time/expiry, cached details and retry/refresh action. **States:** “Cached Result” within seven-day TTL; “Cached Data Expired” after TTL until refreshed. Explain login, uncached search and AI require internet. **Prompt:** Design SafeBite offline states with “Cached Result” and “Cached Data Expired” banners, retrieval time, expiry, cached details and refresh action. Explain that login, uncached search and AI are unavailable offline. Never present expired data as current.

4 Reusable interaction states

Create reusable loading skeletons, empty states, network failure/retry, barcode-not-found, exact status/reason blocks, cached/expired banners, expired-session handling, AI insufficient-evidence state, consent/deletion dialogs, and confirmation dialogs.

5 Figma/Stitch workflow

1. Create pages: Design System, Authentication, Core Scan Flow, Analysis and Evidence, AI Assistant, Health Context, Saved and Profile.
2. Start with Home, Scanner, Product Details, Analysis and Interpretation.
3. Create reusable buttons, fields, status labels, evidence chips, product cards, contaminant rows, citations, banners, empty states and bottom navigation.
4. Use clearly labelled sample data and keep status text identical to the SRS.
5. Prototype Welcome → Authentication → Home → Scan/Search → Product Details → Interpretation.
6. Add paths to evidence, alternatives, comparison, AI, health context, guidance, history, saved products and offline cache.
7. Test with at least five representative users as specified by the SRS usability plan.

6 Handoff to Antigravity and backend developer

For each screen record its Figma frame, button destinations, required endpoint, request/response shape, error states, offline behavior, and SRS requirement IDs. Use mock data while APIs are being implemented. The React Native app calls the backend API; it must not connect directly to Prisma or PostgreSQL. Never put database credentials or AI provider keys in the mobile app.

7 Design-completion checklist

- All 17 SRS features are mapped to screens/states.
- Scan/search-to-interpretation is prototyped and easy to complete.
- Results show contaminant, value/unit, status, evidence level and selected source.
- Missing data, incomparability, no health-context evidence and offline cache are explicit.
- Health-context consent/edit/delete and exposure guidance rules are represented.
- Every interactive-looking control has a defined behavior.
- Figma and backend API contract use matching names and fields.